

JEFFREY BLAY

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PROFESSIONAL SUMMARY

Geospatial data scientist and spatial data engineer completing a Ph.D. in Geospatial Data Science (Dec 2026), with 5+ years building production geospatial applications, ETL pipelines, and machine learning systems over large-scale spatial data. Combines GIS development (Python, ArcPy, QGIS plugins, FastAPI, web mapping) with deep learning and cloud deployment, and a record of turning complex spatial analysis into decision-ready products for government, research, and community stakeholders.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, R

GIS & Remote Sensing: ArcGIS Pro, ArcGIS Online, ArcGIS Dashboards, ArcPy, QGIS (plugin development), Google Earth Engine, GDAL/OGR, GeoPandas, rasterio, spatial data QA/QC

Web Mapping & APIs: FastAPI, REST APIs, Leaflet.js, MapLibre GL, Streamlit, HTML/CSS, JSON

Data Engineering: ETL pipeline design, Apache Airflow, data validation & QA/QC, web scraping, batch processing on HPC

Databases: PostgreSQL/PostGIS, MySQL, SQLite, IBM DB2

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, Feature Engineering, physics-informed neural networks (PINNs), RAG / LLM APIs

Cloud & DevOps: AWS (S3, EC2), Docker, CI/CD, Git, GitHub Actions, Render, Railway, Vercel, GitHub Pages

Analytics & Visualization: Spatial statistics, geostatistics, time-series analysis, Power BI, Tableau, Matplotlib, Seaborn, R Shiny, ArcGIS StoryMaps

PROFESSIONAL EXPERIENCE

Geospatial Research Engineer

Jun 2026 – Present

Remote Sensing & GIS Lab, North Carolina A&T State University — Greensboro, NC

- Developing a QGIS plugin that packages CNN and transformer-based flood-depth models for on-demand inference, making research-grade models usable by non-technical GIS end users.
- Assessing disaster impacts on crop-specific land cover and vulnerable communities using federal datasets and Google Earth Engine to support resilience and risk-mitigation planning.
- Leading a peer-reviewed journal publication on a physics-informed AI framework for urban flood modeling and infrastructure vulnerability assessment.

Geospatial Data Science Research Assistant — NASA Environmental Data Project

Sep 2024 – May 2026

GEMS Institute, North Carolina A&T State University — Greensboro, NC

- Developed and optimized CNN and transformer segmentation models (Swin-UNet, SegFormer, UNet, Attention UNet, UNet++) in PyTorch for urban flood-depth prediction, achieving ~0.19 ft average MAE.
- Designed a physics-informed deep learning framework embedding domain-knowledge constraints, improving physical consistency of predictions by ~84% over conventional deep learning baselines.
- Built geospatial ETL pipelines in Python (rasterio, GDAL, pandas) to ingest, align, and preprocess ~347 million pixel-level records from multi-source raster data for model training.
- Published ezprocess, an open-source Python library (PyPI/GitHub) that cut geospatial data-preparation time by ~85% across machine learning workflows.
- Led a 4-person research team to design and publish a novel flood-depth benchmark dataset for reproducible urban inundation modeling.
- Managed code, experiments, and dependency versioning in Git/GitHub, enabling collaborative development across 2 research teams.

Data Science Research Assistant — NSF Multimodal Data Fusion Project

Jun – Aug 2024

Remote Sensing & GIS Lab, North Carolina A&T State University — Greensboro, NC

- Built ETL workflows in Python (NumPy, pandas) preparing ~700K geospatial point records for flood prediction models, reducing data preparation time by 60%.
- Trained and evaluated Random Forest and XGBoost models using train-test splits, cross-validation, and performance metrics to assess predictive accuracy and reliability.
- Developed automated geospatial QA/QC and validation tests verifying spatial accuracy, attribute consistency, and data integrity prior to public release.
- Designed automated multi-sensor preprocessing pipelines with ArcPy in ArcGIS Pro for a 5-member team publishing a multi-source geospatial dataset.

Graduate Research Assistant

Sep 2023 – May 2024

Remote Sensing & GIS Lab, North Carolina A&T State University — Greensboro, NC

- Built Python web-scraping workflows (BeautifulSoup, SQLite) to curate ~1,000 web images into a labeled training dataset, then developed a PyTorch framework to train and evaluate GoogLeNet and ResNet18 classifiers for automated flood detection in crowdsourced imagery.

GIS & Community Engagement Manager

May – Aug 2023

Urban Resources Initiative — New Haven, CT

- Managed 7 community stakeholder groups, translating competing priorities into weekly project plans and implementation strategies.
- Delivered ArcGIS Pro and QGIS analyses, maps, and visualizations enabling stakeholders to evaluate site-specific options and refine proposed investments.

Research Assistant — NASA Environmental Justice Project

Sep 2022 – May 2023

Hixon Center for Urban Ecology, Yale University — New Haven, CT

- Analyzed multispectral satellite imagery at census-block level to identify communities vulnerable to urban heat island effects, building ArcPy and SQL workflows for land surface temperature data.
- Ran time-series analysis in R (dplyr, ggplot2) to characterize heat and tree-cover disparities across urban communities.

Research Fellow — TRI Research Fellowship
Tropical Resources Institute, Yale University — New Haven, CT

May 2022 – May 2023

- Trained a U-Net regression model on Landsat imagery to predict building composition across metropolitan Ghana, achieving R² of ~0.75 on held-out test data.
- Processed VIIRS nighttime-lights data in Google Earth Engine and applied geostatistical analysis to map electricity infrastructure gaps across built-up areas.
- Deployed an integrated field mapping and monitoring system (ArcGIS Pro, ArcGIS Online, Field Maps, Dashboards) capturing 300+ ground-truth observations and improving collection accuracy by 90%.

Geospatial Data Analyst
National Census Secretariat, Ghana Statistical Service — Accra, Ghana

Sep 2020 – Jun 2021

- Performed enterprise-scale QA/QC on national census datasets, validating coordinate reference systems, boundary alignment, geometry, and attribute integrity across point, line, and polygon data.
- Restructured Supervisory Area boundary datasets, resolving spatial errors, misaligned boundaries, and redundant features to improve national data consistency.
- Produced near-real-time coverage analytics with a 15-member team and identified difficult-to-enumerate areas to direct field resource allocation during census operations.

SELECTED PROJECTS

GridVision MLOps — jeffreyblay.github.io/energy_demand_mlops

2026

Python · FastAPI · PostGIS · Supabase · GitHub Actions · MapLibre GL · Render

- Built an end-to-end geospatial MLOps platform forecasting U.S. electricity demand, with live data ingestion, automated retraining, model gating, and production promotion.
- Developed FastAPI REST services backed by PostGIS, a MapLibre GL 3D visualization layer, and CI/CD via GitHub Actions to a cloud-hosted production deployment.

Quickview Geodata Portal — jeffreyblay.github.io/quickview-geodata-portal

2026

Python · FastAPI · GeoPandas · GDAL · Leaflet.js · Railway · GitHub Pages

- Built an open-source platform for interactive exploration of 5 geospatial file formats loaded directly from URLs, with FastAPI endpoints for ingestion, format conversion, and delivery.
- Deployed cloud-hosted API services on Railway with a Leaflet.js mapping interface on GitHub Pages.

Weather Alert Dashboard — jeffreyblay.github.io/climate-alert-dashboard

2026

Python · FastAPI · JavaScript · Docker · CI/CD · REST API · GitHub Pages

- Built a real-time climate risk platform integrating the National Weather Service API via an automated ETL pipeline across all 50 U.S. states.
- Developed REST services with filtering by state, severity, and event type plus automated risk scoring, deployed as a Dockerized API-first architecture with a JavaScript dashboard.

EDUCATION

Ph.D., Applied Science & Technology — Geospatial Data Science

Expected Dec 2026

North Carolina A&T State University

Relevant Coursework: Machine Learning and Data Mining, Big Data Analytics, Advanced Geospatial Analysis, Neural networks, Statistical Methods, Multivariate Statistics for Engineers

M.E.S., Environmental Science — Environmental Data Science for Urban Sustainability

May 2023

Yale University, School of the Environment

Relevant Coursework: Advanced Remote Sensing, Statistics for Environmental Sciences, Data Exploration & Analysis, Environmental Data Science, Introductory Machine Learning, Urban Landuse Clinic

B.A., Geography with Political Science

May 2019

University of Ghana | GPA: 3.81/4.00

Certifications: IBM Data Engineering Professional Certificate, Coursera (in progress, exp. Sep 2026); Foundations of AI Engineering, CodePath (2026)

HONORS

Outstanding Doctoral Graduate Research Assistant Award, College of Science & Technology, NC A&T State University (2026); Lidar Leader Award — Best Poster, ASPRS & Geo Week Conference (2024).